L1 Cognitive Neurology

Cognitive and Behavioural Neuroscience (CBN)

L1 Memory and Memory Disorders

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EP/Medic/BMS Tutorials - Week 2
MT 2026

1

ESSAY Q: What do memory disorders tell us about the organisation of human memory?

WEEK 2 L1 Cognitive Neurology Tutorials

ESSAY: What do memory disorders tell us about the organisation of human memory?

In this essay, you should explain and critically evaluate the major forms of human memory.

Your goal for this tutorial is to get to an **understanding of how the different forms of memory are defined** and how they **operate together** to support everyday cognition, and how studies of memory disorders have informed our understanding of the organisation of human memory.

In your essay you can use:

- Studies of neuropsychology;
- Clinical case-studies;
- Famous neurological cases;
- Findings from neuroimaging;

To support your arguments about how different forms of memory are organised, how they interact, and the systems that support them. E.g. you could distinguish between short-term/working memory, long-term memory, and the principal subsystems within long-term memory. You can discuss how specific neurological disorders, events, or case-studies of patients have shaped our understanding of when and how these systems fail, and how those distinct patterns of impairment and preserved function have helped reveal the contributions of different brain regions to cognition.

2

Essay

How have studies of memory disorders helped us to understand memory systems in humans?

Memory - 18 times (11)

- S2024 - What have we learned about episodic memory systems from the study of patients with brain damage?
- M2024 - To what extent can human memory be characterized in terms of the storage and retrieval of information?
- S2023 - What have we learned about episodic memory systems from the study of patients with brain damage?
- S2022 - To what extent can human memory be characterized in terms of the storage and retrieval of information?
- M2021 - What have we learned about episodic memory systems from the study of patients with brain damage?
- M2019 – Human memory is a reconstructive rather than reproductive process. Discuss.
- S2018 – 'Human memory acts a reconstructive rather than reproductive process.' Discuss.
- M2018 – How are false memories established? What are the clinical and societal implications?
- S2017 - Why do false memories occur and what impact do they have in legal and other practical contexts?
- M2016 - 'Human memory acts as a reconstructive rather than reproductive process'. Discuss.
- S2015 - Human memory is a reconstructive rather than reproductive process. Discuss.
- M2015 - Discuss evidence that distinct psychological components are linked to different sections of the serial position curve in human memory research.
- S2014 - Why do false memories occur? What are the practical implications?
- M2013 - What do 'memory illusions' tell us about the nature of human memory and what are the practical implications?
- S2012 - To what extent can one characterise human memory in terms of the storage and retrieval of information? What are false memories?
- S2011 - What evidence suggests that people exhibit "false memories"? Why do these false memories arise?
- M2010 - What three psychological processes form the basis of human memory? In your answer give at least one example of an experiment or test where any of these three processes are measured.
- J2009 - What are encoding, storage and retrieval in memory? Give one example of an experiment or test in which one of these processes is measured.
- M2009 - Describe 'working memory' and an experiment to measure it.

3

CBN Specific Exam Essays L1 Cognitive Neurology

1. Discuss amnesia and confabulation. What have we learnt about the brain systems and cognitive mechanisms that might be damaged in each of these phenomena?

4. What kinds of episodic memory disorders have been reported and what do they reveal about underlying cognitive mechanisms and brain systems?

3. What does the human hippocampus do? Discuss using examples from patient studies.

1. How have studies of patients advanced understanding of episodic and semantic memory?

4. Discuss the cognitive mechanisms and brain regions implicated in confabulation.

3. Describe visual associative agnosia and semantic dementia. What is the relationship between these disorders?

2) How have double dissociations of patients with short- and long-term memory deficits contributed to our understanding of human memory processes and systems?

4. What are the disorders of episodic memory that have been described and how have they helped us to understand normal memory functions?

4

Stages of memory

Sensory

ICONIC
ECHOIC
HAPTIC
NOCICEPTIVE
…

Short-term

Long-term

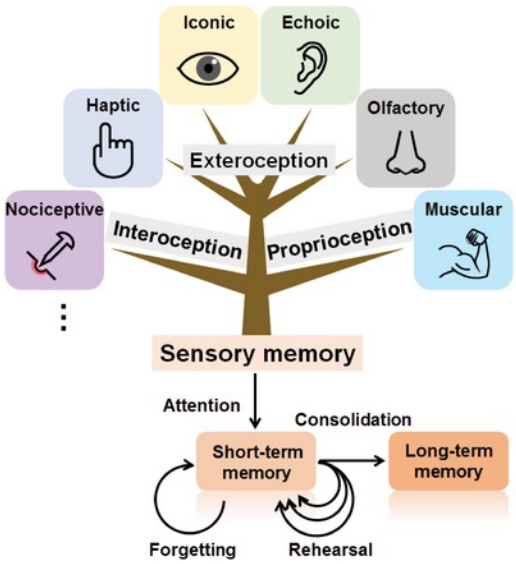
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Stages of memory

Sensory

ICONIC
ECHOIC
HAPTIC
NOCICEPTIVE
…

Coltheart, M. (1980). Iconic memory and visible persistence. *Perception & psychophysics*, *27*, 183-228.

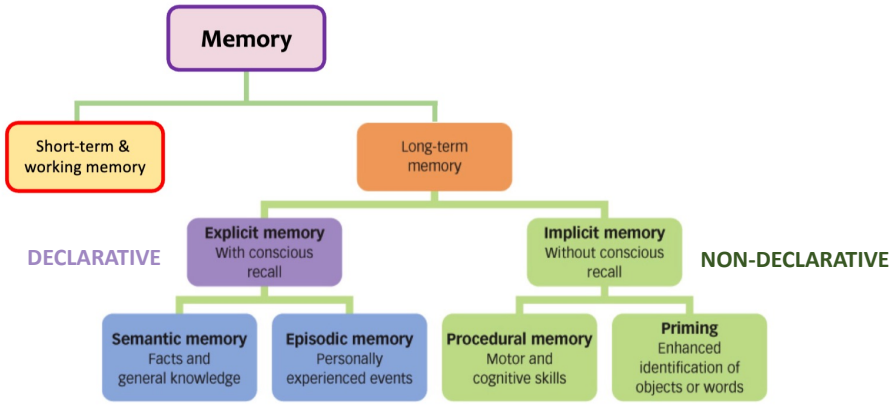


(visual) sensory memory -- can be experimentally studied by using the partial report technique (Coltheart, 1980)

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Different subsystems of memory

Fractionation of memory systems

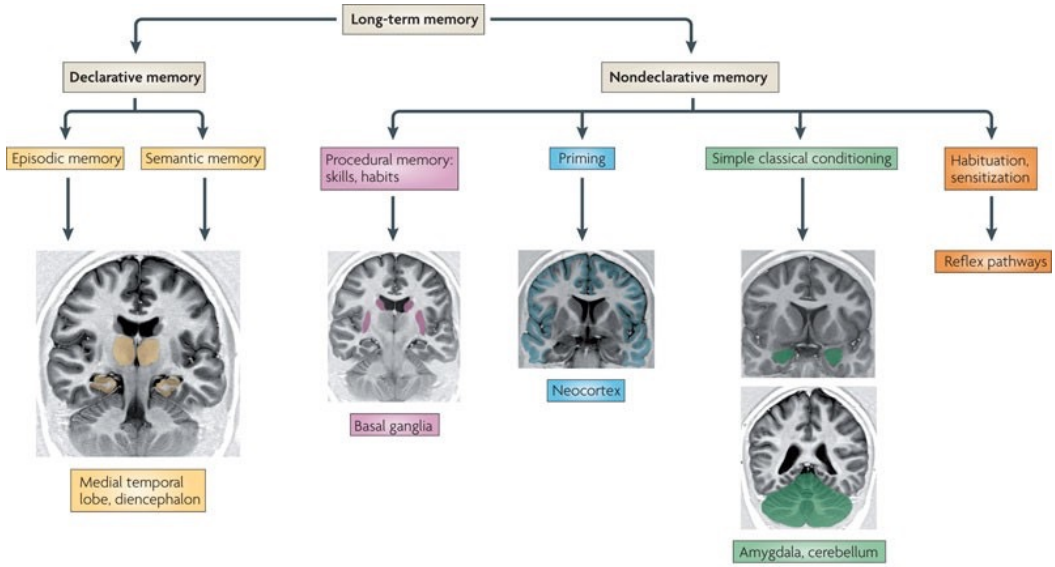


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Different subsystems of memory

Has this taxonomy (Squire et al., 2009) held up?

8



Henke, K. A model for memory systems based on processing modes rather than consciousness. *Nature Review Neuroscience* **11**, 523–532 (2010). https://doi.org/10.1038/nrn2850

9

Q: What distinguishes remembering an event from knowing a fact about that event?

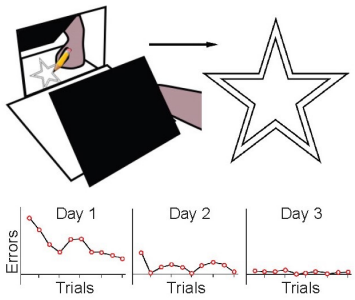
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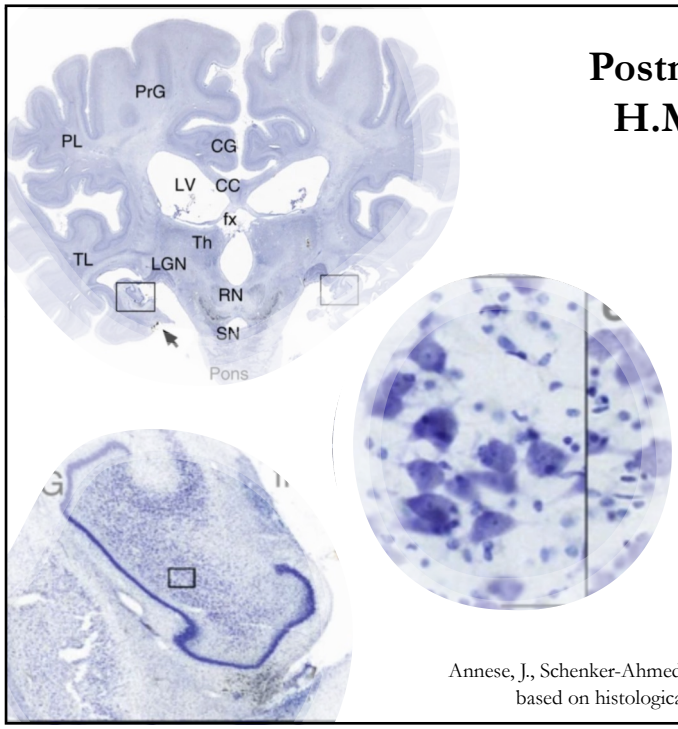
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Q: H.M. improved at mirror drawing but could not remember doing it. Does this mean he could learn but could not remember?





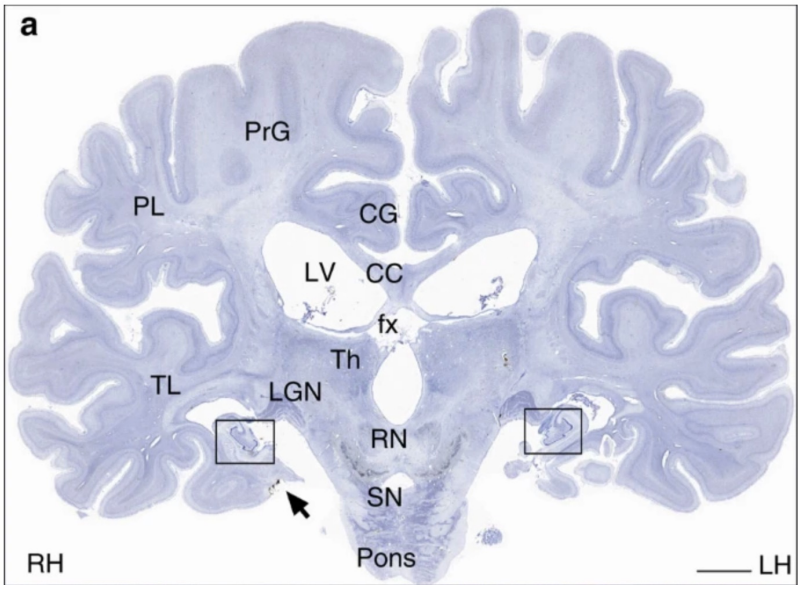
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Postmortem examination of patient H.M.'s brain based on histological sectioning and digital 3D reconstruction

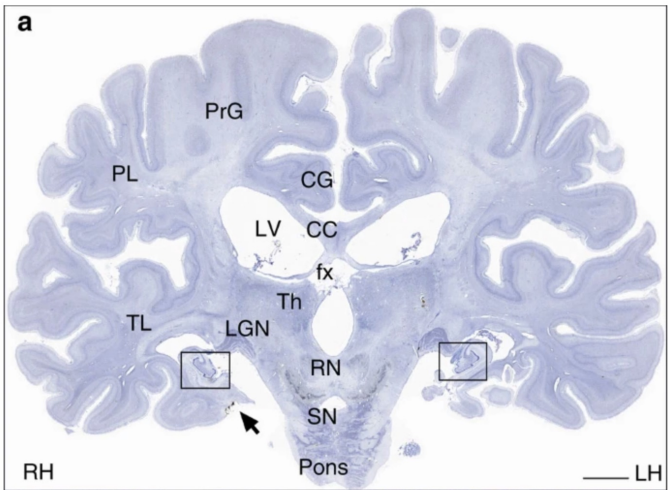
Annese, J., Schenker-Ahmed, N., Bartsch, H. *et al.* Postmortem examination of patient H.M.'s brain based on histological sectioning and digital 3D reconstruction. *Nat Commun* **5**, 3122 (2014).

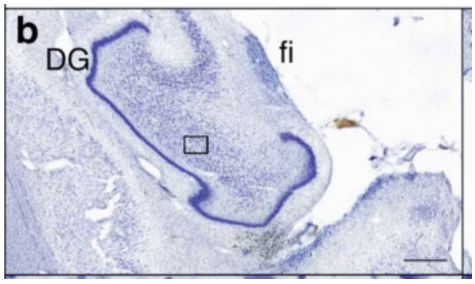
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Annese, J., Schenker-Ahmed, N., Bartsch, H. *et al.* Postmortem examination of patient H.M.'s brain based on histological sectioning and digital 3D reconstruction. *Nat Commun* **5**, 3122 (2014).

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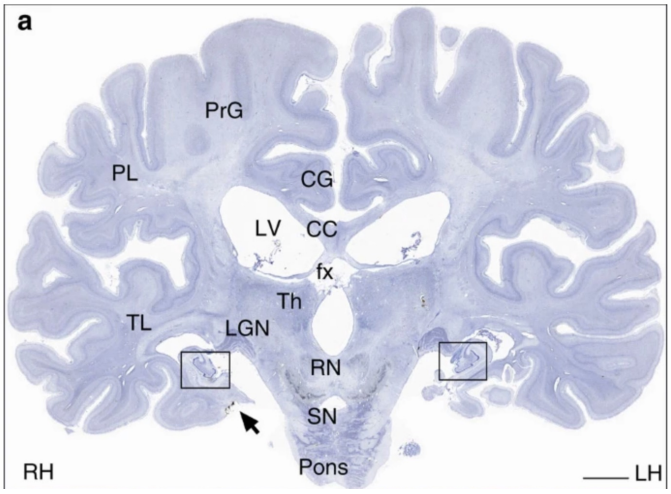


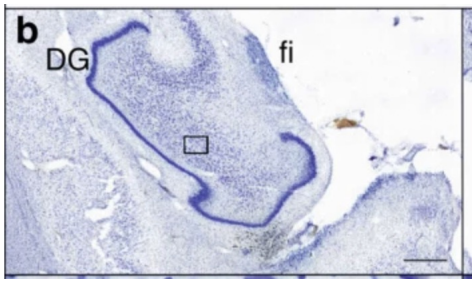


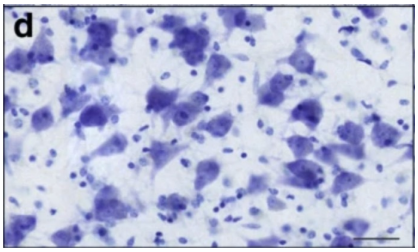
If H.M. retained substantial hippocampal tissue, why was his memory impairment so profound?

Annese, J., Schenker-Ahmed, N., Bartsch, H. *et al.* Postmortem examination of patient H.M.'s brain based on histological sectioning and digital 3D reconstruction. *Nat Commun* **5**, 3122 (2014).

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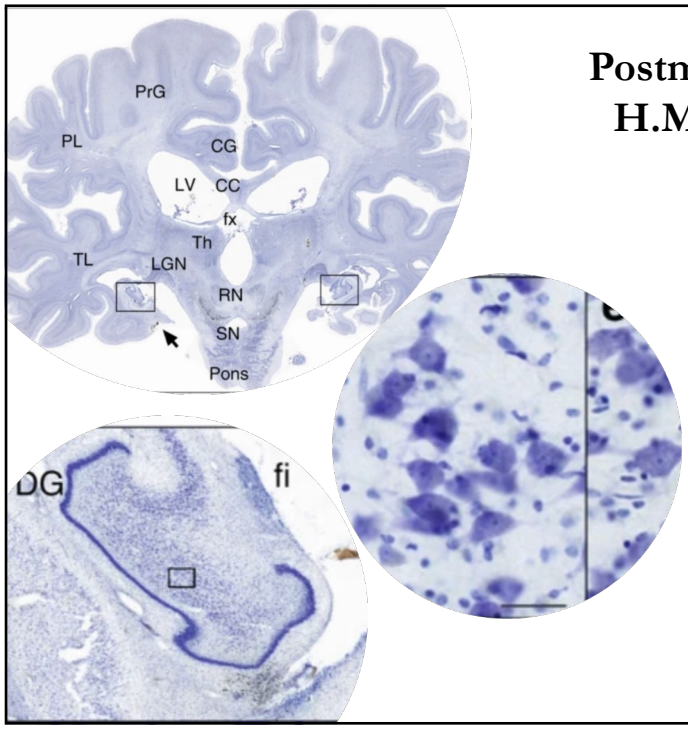






Annese, J., Schenker-Ahmed, N., Bartsch, H. *et al.* Postmortem examination of patient H.M.'s brain based on histological sectioning and digital 3D reconstruction. *Nat Commun* **5**, 3122 (2014).

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Postmortem examination of patient H.M.'s brain based on histological sectioning and digital 3D reconstruction

Key findings:

- The hippocampus was not completely removed. Approximately 2 cm^3 of hippocampal tissue remained in each hemisphere
- The entorhinal cortex was almost entirely removed.
- Other pathology was present, including deep white-matter lesions and a small left orbitofrontal lesion

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Memory depends not only on individual brain structures but also on the integrity of the connections between them. H.M.'s case illustrates both the extraordinary value and the anatomical limitations of single-patient lesion evidence



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Amnesias: anterograde amnesia

- Impaired formation of new long-term memories following neurological injury or disease.
- Typically affects the formation of declarative memories, with relatively preserved immediate memory span and some forms of procedural learning.
- Bilateral hippocampal formation, medial temporal lobe structures and hippocampal-diencephalic circuitry may be affected
- **Patient H.M**. -- developed profound anterograde amnesia following bilateral medial temporal lobe resection for refractory epilepsy.

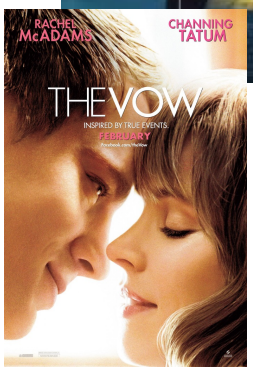


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Amnesias: retrograde amnesia

- Loss of memories acquired *before* the onset of neurological injury or disease. Popular (narratively) with films, but less common than anterograde amnesia in real-life
- May affect autobiographical episodic memories, semantic knowledge or both.
- Often exhibits a temporal gradient (Ribot's law), with recent memories more vulnerable than remote memories.
- Hippocampal-neocortical networks, including medial temporal, lateral temporal and distributed association cortices.
- Clinical examples: Traumatic brain injury, herpes simplex encephalitis and some neurodegenerative disorders.

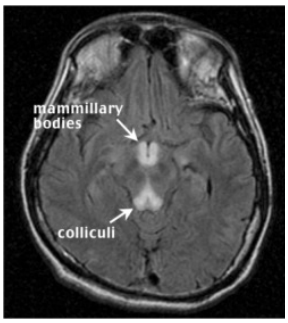




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Amnesias: Korsakoff's (diencephalic amnesia)

- Chronic amnestic syndrome resulting from **thiamine deficiency;**
- Characterised by **profound anterograde amnesia + variable retrograde amnesia**, temporal disorientation and occasional **confabulation**.
- Associated with damage to the mammillary bodies, medial thalamus and hippocampal-diencephalic circuitry, often with additional executive dysfunction.



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Amnesias: Dense amnesia / Transient Global Amnesia

Transient global amnesia, which presents as a single episode of dense amnesia with complete recovery and no features of epilepsy, and has a very low recurrence rate.

Differentiate this from transient epileptic amnesia, which is recurrent

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Amnesias: post-traumatic amnesia

- Impaired continuous memory formation following traumatic brain injury, often accompanied by confusion and disorientation.
- Patients may have anterograde amnesia during recovery and retrograde amnesia for events preceding the injury.
- Duration of PTA is an important clinical indicator of traumatic brain injury severity.
- Neuroanatomy: Variable; may involve diffuse axonal injury, frontotemporal structures and distributed attention-memory networks.

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Amnesias: dissociative amnesias

- Amnesias of a psychological nature, as in personality disorders; e.g. dissociative identity disorder
- May involve circumscribed memory gaps or more extensive autobiographical memory loss, sometimes following severe psychological stress.
- New learning and non-autobiographical memory functions remain relatively preserved.
- Neurobiology: No consistent focal structural lesion; proposed mechanisms involve altered functional interactions between prefrontal, limbic and autobiographical memory networks.
- Clinical presentation: Loss of recollection for significant personal events, sometimes accompanied by dissociative fugue

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Amnestic syndrome	Characteristic clinical features	Principal aetiology
Korsakoff syndrome (diencephalic amnesia)	Severe anterograde amnesia, variable retrograde amnesia, temporal disorientation and occasional confabulation	Thiamine deficiency; mammillary bodies, medial thalamus and mammillothalamic tract
Transient epileptic amnesia (TEA)	Brief, recurrent episodes of amnesia, often on waking; may be associated with accelerated long-term forgetting and autobiographical memory loss	Temporal lobe epilepsy, frequently involving medial temporal networks
Accelerated long-term forgetting (ALF)	Apparently adequate initial learning and short-delay retention, followed by disproportionate forgetting over days or weeks	Associated with temporal lobe epilepsy and other neurological disorders; mechanisms may involve impaired longer-term consolidation or retention
Developmental amnesia	Severe episodic memory impairment following early hippocampal injury, with relatively preserved acquisition of semantic knowledge	Bilateral hippocampal damage, often following childhood hypoxic–ischaemic injury
Post-encephalitic amnesia	Severe anterograde and variable retrograde amnesia following encephalitis	Herpes simplex encephalitis frequently affects medial temporal and orbitofrontal structures; autoimmune limbic encephalitis may predominantly affect medial temporal structures
Basal forebrain amnesia	Impaired new learning, sometimes with confabulation and executive dysfunction	Basal forebrain injury, including following anterior communicating artery aneurysm rupture
Transient amnesia associated with migraine	Temporary memory disturbance occurring in association with migraine; diagnosis requires exclusion of TGA, epilepsy and other causes	Mechanism uncertain; not a distinct, consistently defined amnestic syndrome

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Amnesia

- **Medial Temporal Lobe** – pt H.M. and R.B.
  Mainly anterograde amnesia, but some retrograde amnesia
- **Mamillary bodies** – Korsakoff's Syndrome
  Anterograde and retrograde amnesia a/w alcoholism
- **Learning in amnesia** – are episodic, semantic, and procedural information different?

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Paramnesia (distortions of memory)

1. Distortions of recall
a) Cryptamnesia
b) False memory
c) Pseudologia fantastica
d) Confabulation
e) Munchausen's syndrome
2. Distortions of recognition.
a) Hyperamnesia
b) Déjà vu



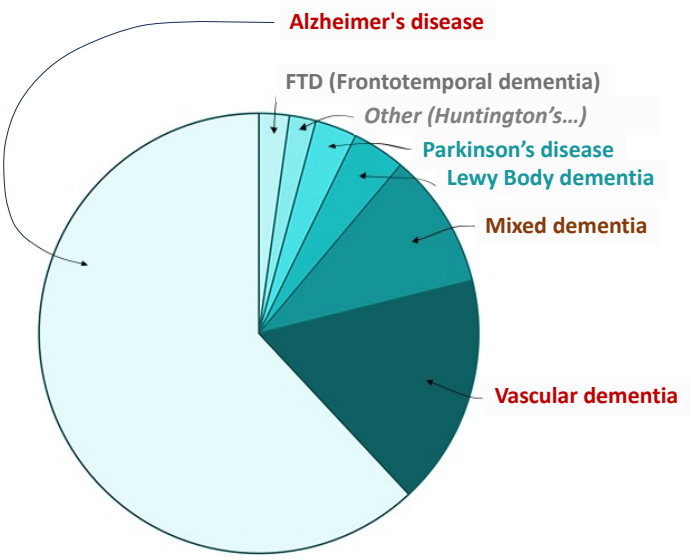
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What is anomia?

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Dementia



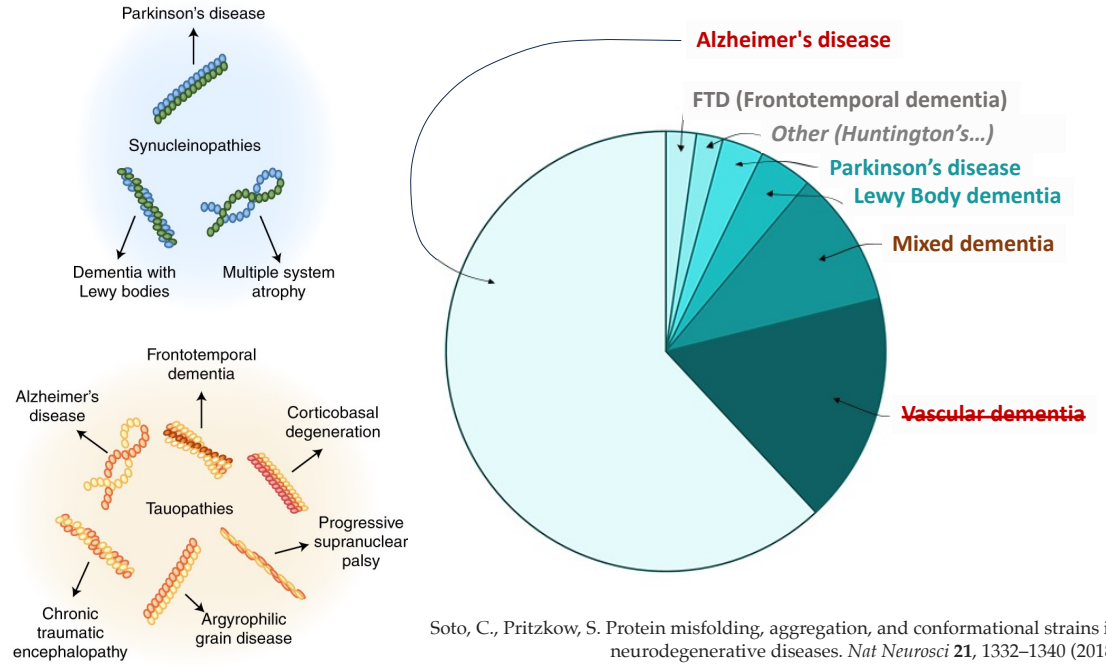


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Dementia

- Duration – ICD specifies minimum of 6 month duration of cognitive deficits
- Course may vary – stabilized cognitive dysfunction or progressive
- On examination underlying medical condition may be manifest, often may have age-related or no physical signs
- Screening – History and MSE, MMSE, HMSE

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Soto, C., Pritzkow, S. Protein misfolding, aggregation, and conformational strains in neurodegenerative diseases. *Nat Neurosci* **21**, 1332–1340 (2018)

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FTD

- Male predominant
- Atrophy in the frontal and temporal lobes
- Variants –PPA (agrammatic, semantic, logopenic), behavioural variant
- TDP-43, FUS, Tauopathy
- Pick bodies (intraneuronal argentophilic inclusions) and Pick cells (swollen neurons)

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Semantic Dementia

- Initially a progressive speech disturbance – Impaired comprehension;
- **Anomia**;
- semantic parapahasias – Normal fluency - effortless speech lacking
- meaning & information – Normal repetition
- Incomplete awareness, particularly impaired comprehension
- Dyslexia / dysgraphia
- Read and write phonetically
- Verbal fluency tests – category fluency > letter fluency
- MRI- temporal atrophy, L > R; more anteriorly than AD

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'The search for the neuroanatomical locus of semantic memory has simultaneously led us nowhere and everywhere...'

Review Article | Published: December 2007

Where do you know what you know? The representation of semantic knowledge in the human brain

Karalyn Patterson, Peter J. Nestor & Timothy T. Rogers

Nature Reviews Neuroscience **8**, 976–987 (2007) | Cite this article

Save article

Neuroimaging studies of semantic memory: inferring "how" from "where"

Sharon L Thompson-Schill

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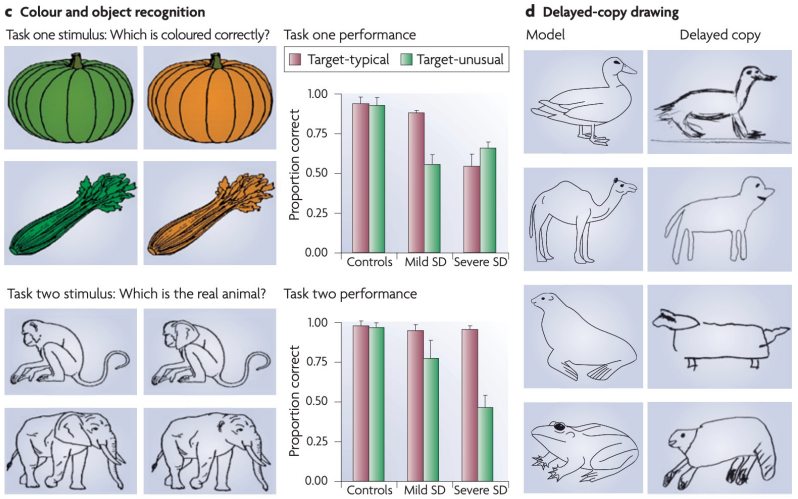


Figure 2 | **Examples of impaired performance on semantic tasks in patients with semantic dementia.** This figure illustrates the cross-modal nature of the impairment and the preservation of general relative to specific information in semantic dementia (SD). **a** | Accuracy at discriminating targets from distractors in a picture-categorization task for four groups of participants: healthy controls and patients with mild, moderate and severe SD[72]. Participants viewed a category label followed by a colour photograph and were asked whether the picture matched the label. Labels were either general (for example, 'animal'), basic-level (for example, 'dog') or specific (for example, 'Labrador'). **b** | Picture-naming responses for one SD patient who was assessed longitudinally[68]. + denotes a correct response. **c** | Examples of stimuli and performance (for controls, mild and severe SD patients) on two recognition tasks. In the first task, participants judged which of two items was coloured correctly. Patients with milder SD performed well when the

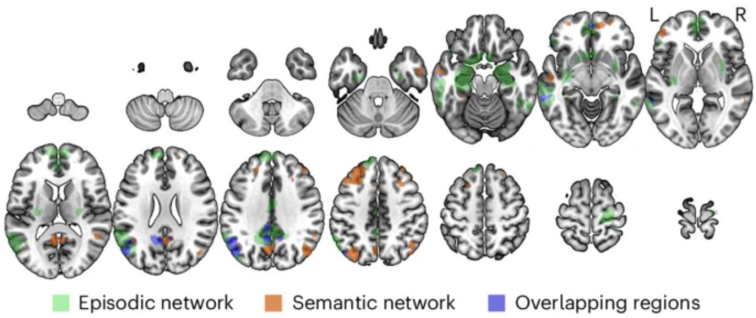
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Registered Report | Published: 27 January 2026

Neural activations and representations during episodic versus semantic memory retrieval

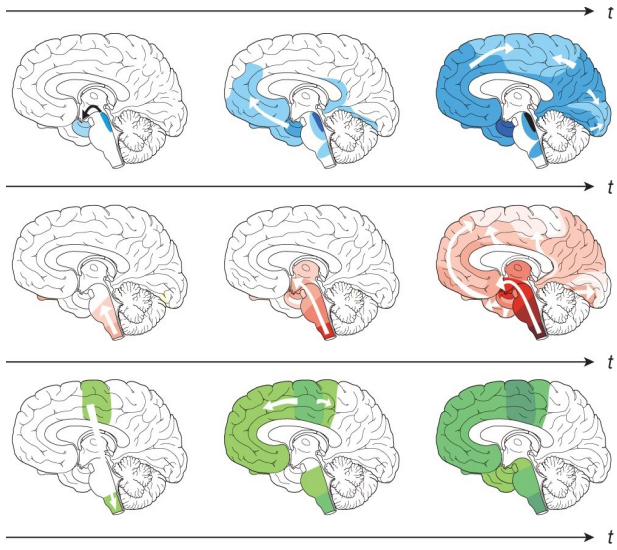
Roni Tibon ✉, Andrea Greve, Gina Humphreys, Jörn Alexander Quent & Richard Henson

Nature Human Behaviour **10**, 803–821 (2026) | Cite this article



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Neurodegenerative neuropathology



Jucker, M., Walker, L. Self-propagation of pathogenic protein aggregates in neurodegenerative diseases. *Nature* **501**, 45–51 (2013).

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Vascular Dementia

- Male, advanced age, hypertension, or other vascular risk factors
- Stepwise or gradual, focal neurologic symptoms, Carotid bruit, fundoscopic abnormalities, and enlarged cardiac chambers
- Hyperintensities and focal atrophy
- Control of risk factors - hypertension, smoking, diabetes, hypercholesterolemia,
  and hyperlipidemia

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Testing memory

How do we actually assess memory, and how does this show that different types of memory are distinct?

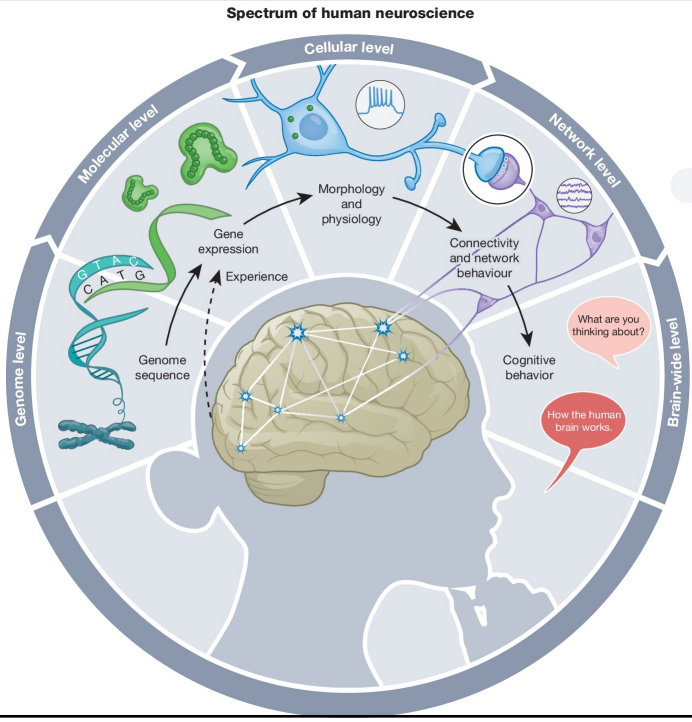
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Memory

What constitutes confabulation?

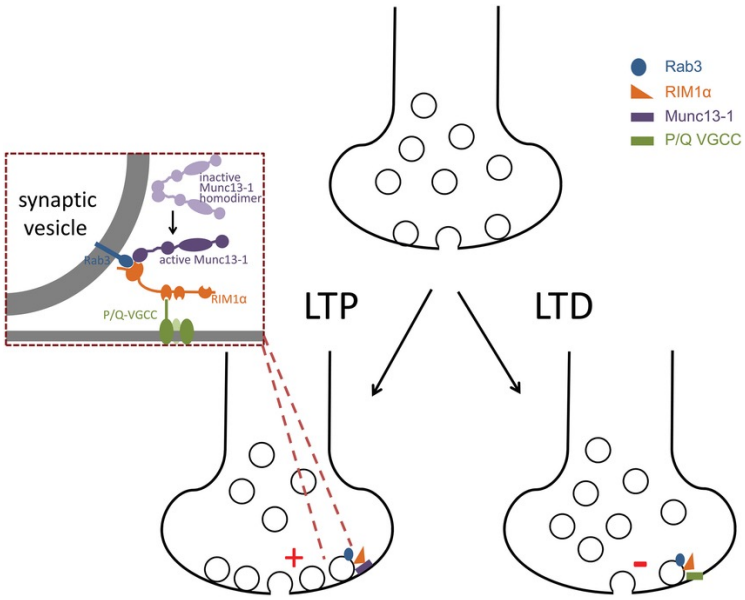
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Memory
across scales



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Cellular memory
synaptic plasticity



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What is Hebb's postulate?

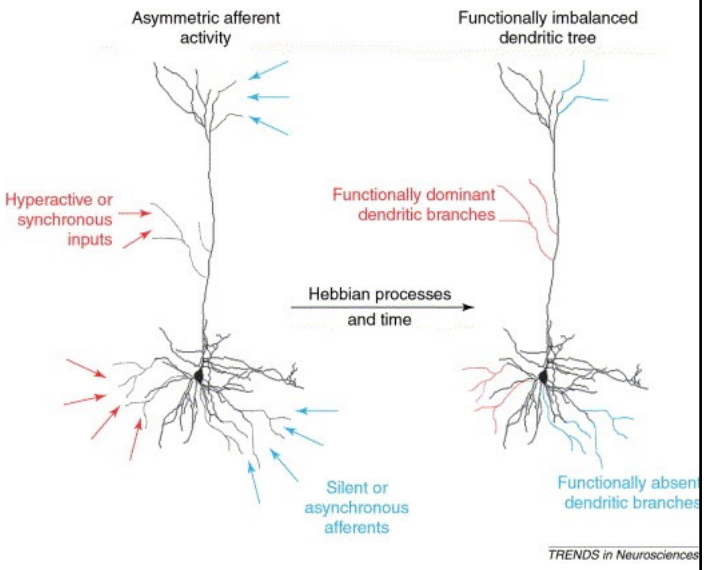
"*cells that fire together, wire together*"

Synaptic strengthening & LTP:

• When **two neurons repeatedly activate together**, their synaptic connection is strengthened, forming the basis of **long-term potentiation (LTP)**. This mechanism is crucial for associative learning and memory storage, particularly in the hippocampus.

Cell assemblies & memory networks:
• Hebb proposed that memories are stored in **distributed neural networks** == **cell assemblies**.

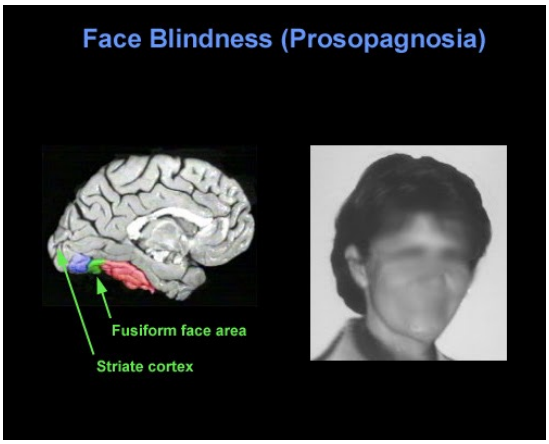
• Activation of part of the assembly can reactivate the whole network, enabling memory retrieval.



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Memory of faces

Prosopagnosia



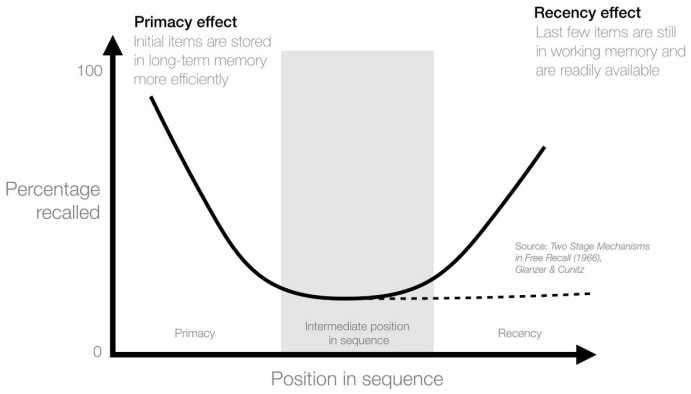


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Sensory memory

Sensory memory: milliseconds to seconds
Short-term memory: seconds to minutes

- Reason for forgetting: decay v. interference (usu interference)
- *Order* matters: serial pattern effect
  - **Primacy and Recency effects** – we are better at remembering things in the beginning and ending of a list
    - **Primacy**: transfer occurs from short to long-term memory through repetition
    - **Recency**: retention in short-term memory



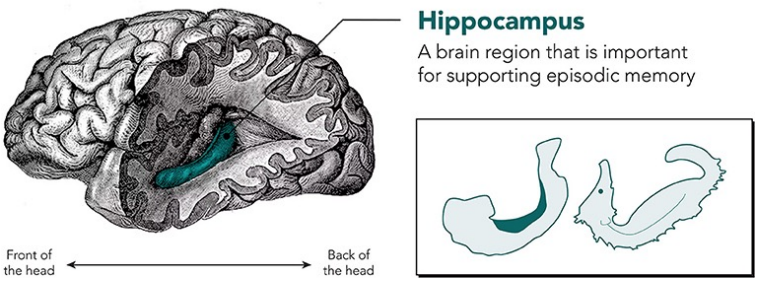
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Episodic memory

Q: what is episodic memory?

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Episodic memory
hippocampus



- **Episodic memory encoding** occurs primarily in the **medial temporal lobe** , especially the **hippocampus, entorhinal cortex, and perirhinal cortex**.
- The **prefrontal cortex** plays a role in retrieval, organisation, and contextual recall.
- **Hippocampal damage (e.g., patient HM)** leads to anterograde amnesia, but remote memories may be spared due to *systems consolidation* (transfer to neocortex).
- Some amnesic patients (e.g., HM, KC) **retain remote episodic and semantic memories** because older memories rely less on the hippocampus over time and become cortically distributed.

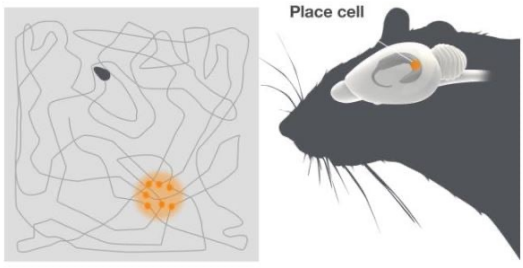
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Animal models of memory
Spatial memory: place cells and grid cells

- **Place Cells:** Located in the hippocampus (especially in the CA1 and CA3 subfields).
- **Grid Cells:** Found in the entorhinal cortex (medial entorhinal cortex - MEC), which is the main input to the hippocampus

- The **hippocampus** plays a crucial role in various stages of learning, including memory encoding, consolidation, and recall, particularly in the context of spatial learning.
  - **Place cells** in the hippocampus exhibit distinct firing patterns in different locations within an environment..

  … forming a cognitive representation of the explored space. When learning goals within the environment, these place cells shift their firing patterns to align with the learned

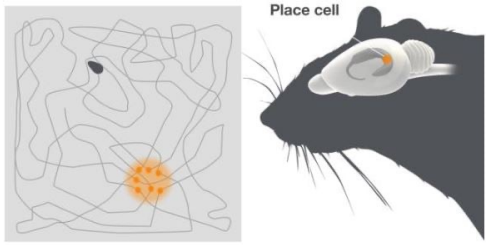


48

Animal models of memory

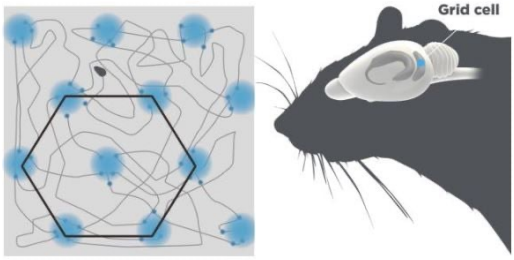
Spatial memory: place cells and grid cells

- **Place Cells:** Located in the hippocampus (especially in the CA1 and CA3 subfields).
- **Grid Cells:** Found in the entorhinal cortex (medial entorhinal cortex - MEC), which is the main input to the hippocampus



2014 Nobel Prize in Physiology (O'Keefe, Moser)

Major input to hippocampus = entorhinal cortex

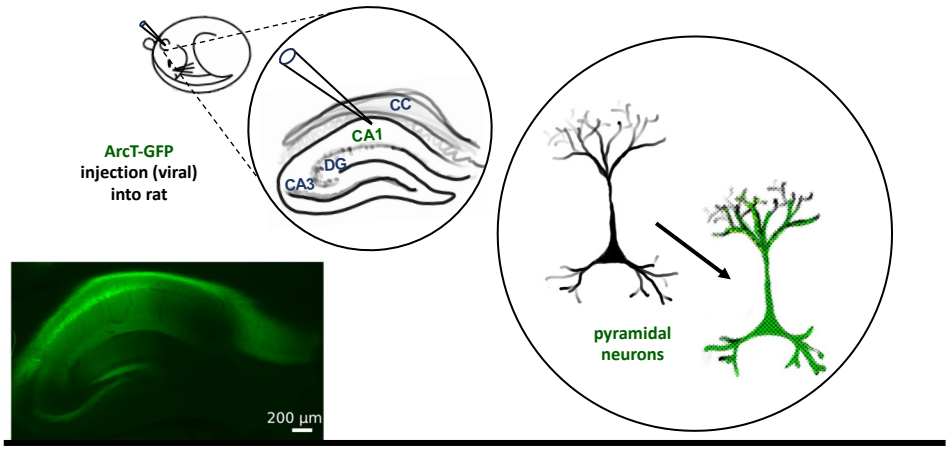


49

Animal models of memory

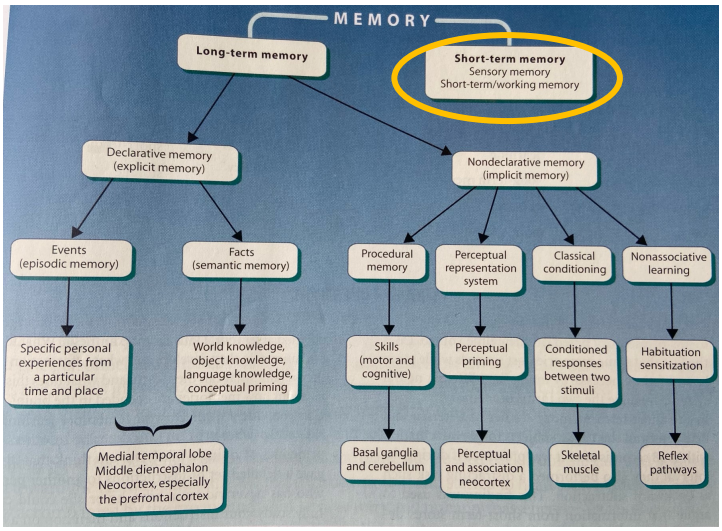
Methods

- These rats had surgery, injected with a virus to express **Archaerhodopsin-GFP** in the CA1
- Rats then had electrodes positioned in **CA1 (hippocampus)**
- this surgical process aimed to allow **real-time decoding of neuronal activity and optogenetic stimulation**



50

What is working memory?



51

Working memory

1970s..

- Baddeley et al proposed the first variant of working memory – information that can be acted on and processed
- Somehow, this will lead to long-term memory if retained
- Pts with amnesia help explain how this works
  Shallice and Warrington – pt w/ L perisylvian damaged reduced digit span to 2, but could make long-term memories

52

Working memory

Phonological loop (left supramarginal gyrus, and L premotor region)
Visuospatial sketchpad (parieto-occipital region of both hemispheres)

- Baddeley et al proposed the first variant of working memory – information that can be acted on and processed

Baddeley & Hitch (1974) Working Memory Model
- Central Executive: Attention control, task-switching (prefrontal cortex)
- Phonological Loop: Verbal STM (left temporoparietal regions)
- Visuospatial Sketchpad: Spatial memory (occipital & parietal lobes)
- Episodic Buffer: Integration with LTM (later – 2000s)

Experimental Evidence
- Dual-task studies (e.g., articulatory suppression)
- n-back task measuring working memory capacity
- Neuroimaging: Activation in **dorsolateral prefrontal cortex**

53

Cognitive ability

Measurement

'Global' Cognitive Ability

MoCA (Montreal Cognitive Assessment)

Famously used by this guy



as evidence of cognitive prowess…

54

Cognitive ability

Measurement

‘Global’ Cognitive Ability

Donald Trump has “incredible genes” and could have lived to 200 years old if his diet had been better in recent years, the White House presidential physician said.

Dr Ronny Jackson, who performed Mr Trump’s first medical exam as president, declared him “fit for duty” physically and mentally.

As well as a thorough physical, the US president asked to be administered a mental test and was given the Montreal Cognitive Assessment. Dr Jackson performed the “difficult” cognitive test and Mr Trump scored 30 out of 30.

55

Cognitive ability

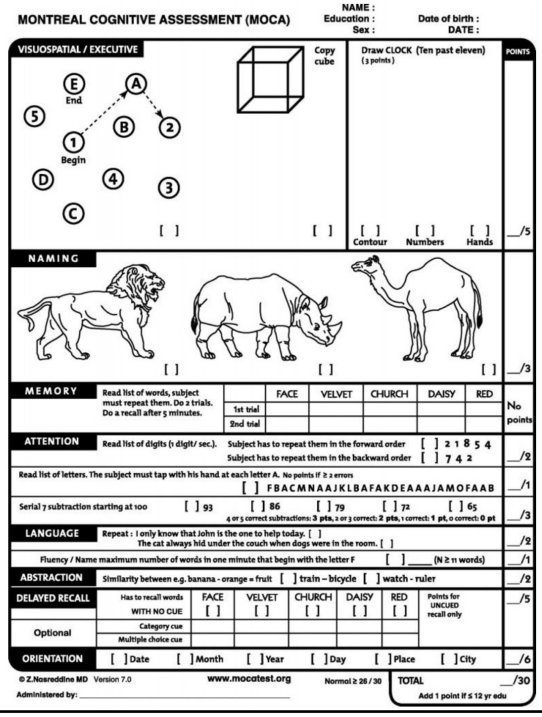
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56

‘Global’ Cognitive Ability

MoCA (Montreal Cognitive Assessment)

Famously used by this guy

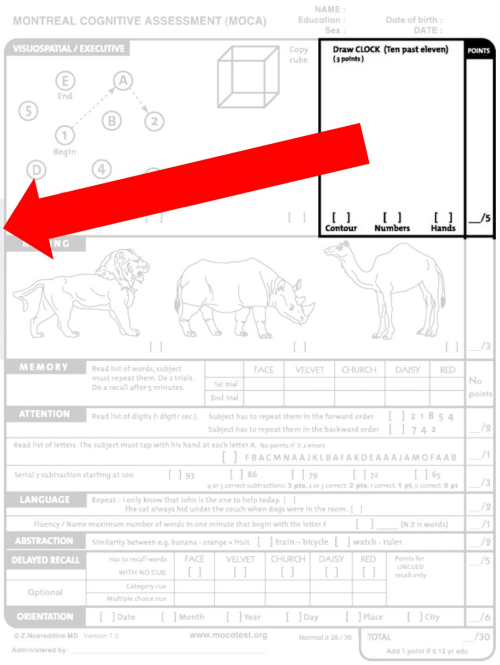


as evidence of cognitive prowess…

Alzheimer’s patients may omit numbers, misplace hands, draw a distorted shape, or fail to complete the clock entirely— why?

Dr Ronny Jackson, who performed Mr Trump's first medical exam as president, declared him “fit for duty” physically and mentally.

As well as a thorough physical, the US president asked to be administered a mental test and was given the Montreal Cognitive Assessment. Dr Jackson performed the “difficult” cognitive test and Mr Trump scored 30 out of 30.



57

Q: Other than the MoCA, can you think of another widely used cognitive test used by clinicians?

58

'Global' Cognitive Ability

MMSE – Mini-Mental State Examination

Mini-Mental State Examination (MMSE)

Patient's Name: _____ Date: _____

Instructions: *Ask the questions in the order listed. Score one point for each correct response within each question or activity.*

Maximum Score	Patient's Score	Questions
5		"What is the year? Season? Date? Day of the week? Month?"
5		"Where are we now: State? County? Town/city? Hospital? Floor?"
3		The examiner names three unrelated objects clearly and slowly, then asks the patient to name all three of them. The patient's response is used for scoring. The examiner repeats them until patient learns all of them, if possible. Number of trials: _____
5		"I would like you to count backward from 100 by sevens." (93, 86, 79, 72, 65, ...) Stop after five answers. Alternative: "Spell WORLD backwards." (D-L-R-O-W)
3		"Earlier I told you the names of three things. Can you tell me what those were?"
2		Show the patient two simple objects, such as a wristwatch and a pencil, and ask the patient to name them.
1		"Repeat the phrase: 'No ifs, ands, or buts.'"
3		"Take the paper in your right hand, fold it in half, and put it on the floor." (The examiner gives the patient a piece of blank paper.)
1		"Please read this and do what it says." (Written instruction is "Close your eyes.")
1		"Make up and write a sentence about anything." (This sentence must contain a noun and a verb.)
1		"Please copy this picture." (The examiner gives the patient a blank piece of paper and asks him/her to draw the symbol below. All 10 angles must be present and two must intersect.) 
30		TOTAL

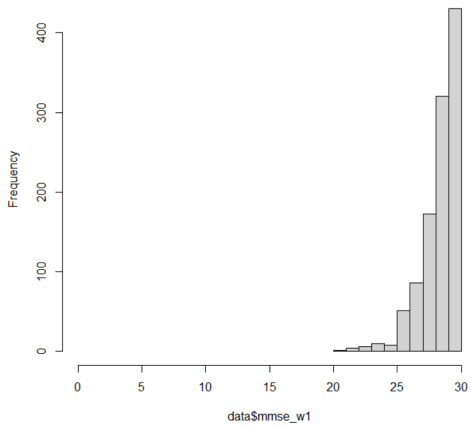
A widely-used screener for 'grading the cognitive state of patients for the clinician'.

Folstein, Folstein & McHugh (1975) *J Psychiatr Res* 12, 189-198

59

'Global' Cognitive Ability

MMSE – Mini-Mental State Examination



A widely-used screener for 'grading the cognitive state of patients for the clinician'.

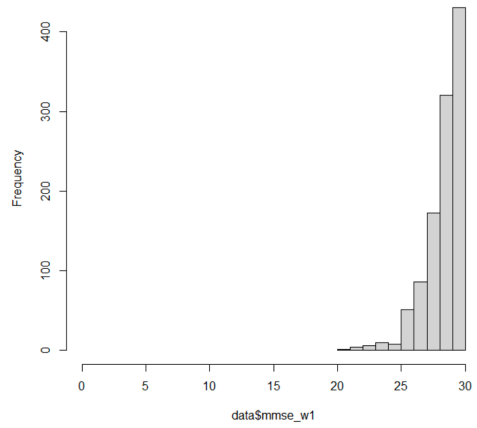
In general these tests (MMSE, MoCA and others) are useful for ascertaining cognitive function among patient groups in clinic and research settings (lesions, dementia etc).

Is this a useful test for assessing memory in 'healthy' patients – aka so called 'normative' cognitive ageing?

60

‘Global’ Cognitive Ability

MMSE – Mini-Mental State Examination



A widely-used screener for ‘grading the cognitive state of patients for the clinician’.

In general these tests (MMSE, MoCA and others) are useful for ascertaining cognitive function among patient groups in clinic and research settings (lesions, dementia etc).

Generally not so helpful within the normal range.

- Ceiling effects
- Insufficient detail for cognitive domains

61

Mini-Cog

1. To administer the Mini-Cog Assessment Instrument

Explain to the person that they should listen carefully to and remember three unrelated words. The three words are then spoken to them, and the person is asked to repeat the words.

Ask the person to mark, on a circle, the hours of a clock then draw the minute and hour hands to show 11:10.

Ask the person to repeat again the three words that they were given at the start.

2. To interpret the Mini-Cog Assessment Instrument

Give 1 point for each word the person correctly remembers after the clock drawing test. Consider the clock drawing test to be normal if all numbers are present in the correct sequence and position, and the hands display the requested time.

- Interpret a score of zero as suggestive of dementia.
- Interpret scores of 1 or 2 with *abnormal* clock drawing test as suggestive of dementia.
- Interpret scores of 1 or 2 with *normal* clock drawing test as a negative screen for dementia.
- Interpret a score of 3 as a negative screen for dementia.

62

Mini-ACE

MINI – ADDENBROOKE'S COGNITIVE EXAMINATION
UK Version A (2014)

Name: Date of Birth: Hospital No. or Address:	Date of testing: ____/___/___ Tester's name:_____ Age at leaving full-time education:_____ Occupation: _____ Handedness: _____

ATTENTION

➢ Ask: What is the	Day _____	Date _____	Month _____	Year _____	**Attention** [Score 0-4] ☐

MEMORY

➢ Tell: "I'm going to give you a name and address and I'd like you to repeat the name and address after me. So you have a chance to learn, we'll be doing that 3 times. I'll ask you the name and address later."

Score only the third trial.

Memory
[Score 0 – 7]
☐

	1st Trial	*2nd Trial*	*3rd Trial*
Harry Barnes	_____ _____	_____ _____	_____ _____
73 Orchard Close	___ _____ _____	___ _____ _____	___ _____ _____
Kingsbridge	_____	_____	_____
Devon	_____	_____	_____

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Appendix 1: Examples of commonly used neuropsychological tests / batteries within memory services

Domain	Standard	Higher Functioning	Non-English Speaking Background
Premorbid optimal ability	TOPF / WTAR / NART / Spot the Word	TOPF / WTAR / NART / Spot the Word	Educational and occupational background Best subtest score e.g. WAIS-IV Picture Completion
Intellectual Functions		WAIS-IV Vocabulary WAIS-IV Similarities WAIS-IV Block Design WAIS-IV Matrix Reasoning	Raven's Advanced Progressive Matrices / Raven's Coloured Progressive Matrices
Memory	RBANS List Learning RBANS Story Recall RBANS Figure Recall	BMIPB II / WMS-IV Logical Memory (& recognition) WMS-IV Visual Reproduction (& recognition) CVLT-III (Short Form)	KBNA List Learning *(translated)* WMS-IV Visual Reproduction KBNA Picture Recognition
Language	RBANS Picture Naming RBANS Fluency	Graded Naming Test / Boston Naming Test	KBNA Picture Naming / Oldfield Naming Test *(qualitative only)*
Visuospatial and perceptual functions	RBANS Figure Copy RBANS Line Orientation	VOSP Position Discrimination VOSP Incomplete Letters Rey Figure	VOSP Position Discrimination VOSP Incomplete Letters *(if letters are known)* WAIS-IV Block Design
Attention and Processing Speed	RBANS Orientation RBANS Digit Span RBANS Coding	WAIS-IV Digit Span WAIS-IV Coding	KBNA Orientation WAIS-IV Digit Span *(if numbers are known)* WAIS-IV Coding *(if numbers are known)* Colour-Trails Part 1 *(if numbers are known)*
Executive Functions	DKEFS Verbal Fluency DKEFS Colour-Word Test DKEFS Trail Making	DKEFS Verbal Fluency DKEFS Colour-Word Test DKEFS Trail Making Hayling Sentence Completion Test	Colour-Trails Part 2 *(if numbers are known)* Modified WCST or Weigl Brixton Spatial Anticipation Test DKEFS Category Fluency and Category Switching
Emotional well-being	PHQ-9 & GAD-7 / HADS	PHQ-9 & GAD-7 / HADS	DISCS / Visual Scale

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BRISTOL ACTIVITIES OF DAILY LIVING SCALE

Date…… *LABEL*

Score ☐

This questionnaire is designed to reveal the everyday ability of people who have memory difficulties of one form or another.

For each activity (No. 1 - 20), statements a - e refer to a different level of ability.
Thinking of the last 2 weeks, tick the box that represents your relative's/friend's AVERAGE ability.

1. PREPARING FOOD		a) Selects and prepares food as required
		b) Able to prepare food if ingredients set out
		c) Can prepare food if prompted step by step
		d) Unable to prepare food even with prompting and supervision
		e) Not applicable
2. EATING		a) Eats appropriately using correct cutlery
		b) Eats appropriately if food made manageable and /or uses spoon
		c) Uses fingers to eat food
		d) Needs to be fed
		e) Not applicable
3. PREPARING DRINK		a) Selects and prepares drinks as required
		b) Can prepare drinks if ingredients left available
		c) Can prepare drinks if prompted step by step
		d) Unable to make a drink even with prompting and supervision

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Symptoms in patients with memory disorders

- Alongside memory impairment…
  - **Aphasia**
  - **Apraxia**
  - **Agnosia**
  - **Executive function impairment** - ability to think abstractly and plan such activities as organising, shopping and maintaining a home

66

Delirium (as distinct from memory disorders)

- Often refers to varieties of acute organic reaction, >40% of hospitalized elderly
- Commonly acute onset, global cognitive dysfunction, often metabolic or substance
- Prominent disturbances in alertness, as well as confusion and a short, fluctuating course, agitation or stupor, fear, emotional lability, hallucinations, delusions, and disturbed psychomotor activity

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Characteristics of Delirium

- Diffuse cognitive deficits – attention, orientation, memory, visuo-constructional ability, executive functions
- Temporal course – acute / abrupt
- Perceptual alterations, thought and mood
- Sleep-wake disturbances
- Psychomotor activity

68

What to rule out?

There are many factors that may influence or impair cognition and these should be considered when assessing someone who presents with cognitive problems including:

- Cerebrovascular events.
- Hypothyroidism.
- Hyperparathyroidism, hypoparathyroidism.
- Hypoperfusion - eg, heart failure.
- Head trauma, including recurrent trauma of having been a boxer.
- Deficiencies of folate, vitamin B12 and vitamin B6 are associated with neurological and psychological dysfunction and are potential factors for cognitive impairment and the development of dementia in the elderly.[6]
- Open heart surgery with cardiopulmonary bypass.
- Medication use, especially sedatives.
- Hepatic impairment.
- Sleep disorders - eg, obstructive sleep apnoea.
- Depression.
- Psychological stress.
- Drug or alcohol abuse.
- Toxins, infections, metabolic (eg, hypoglycaemia) and structural causes.

69

Case 1

Mrs A, formerly an industrious business woman, came to psychiatric attention at the age of 54 years. Over the preceding months she had become increasingly apathetic, lacking in motivation and self-care. If left to her own devices she would spend all day doing nothing. There was an episode of urinary incontinence, which she shrugged off saying that she “could not be bothered” to go to the bathroom. There was no relevant previous medical or family history. She was initially suspected of being depressed. However, her lack of response to therapy and continuing decline raised the suspicion of a neurodegenerative disorder. Two years after onset of her symptoms she was referred to a specialist early onset dementia clinic. Neurological examination was normal aside from ‘frontal’ primitive reflexes. Neuropsychological test performance was characterised by economy of effort and poor persistence on tasks. She showed profound impairment on frontal executive tests, with striking perseveration of both verbal and motor responses. There was no primary aphasia, perceptual or spatial disorder and she was oriented in time and place. A SPECT scan showed bilateral frontal hypoperfusion. [redacted]

70

Case 1

Over the ensuing years speech output became progressively more attenuated, echolalic and perseverative , culminating in total mutism. She became gluttonous, with a fondness for sweet foods leading to obesity. In the latter stages, she failed to eat, attributable to her profound apathy, resulting in dramatic weight loss. Her affect was blunted and unconcerned. Mild parkinsonian signs emerged late in the disease course. She died of bronchopneumonia 7 years after onset of symptoms. Post-mortem examination of the brain demonstrated severe, circumscribed atrophy, particularly affecting dorsolateral frontal regions, and associated with tau (Pick-type) histopathology.

71

Case 1 = FTD

increasingly apathetic,

lacking in motivation

Points of note

- The early clinical features suggested affective disorder
- The clinical presentation of FTD may be of apathy and inertia rather than frank social disinhibtion

"could not be bothered"

normal aside from 'frontal' primitive reflexes.

profound impairment on frontal executive tests, with striking perseveration of both verbal and motor responses. There was no primary aphasia, perceptual or spatial

disorder and she was oriented in time and place. A SPECT scan showed bilateral frontal hypoperfusion. A diagnosis of FTD was made.

Post-mortem examination of the brain demonstrated severe, circumscribed atrophy, particularly affecting dorsolateral frontal regions, and associated with tau (Pick-type) histopathology.

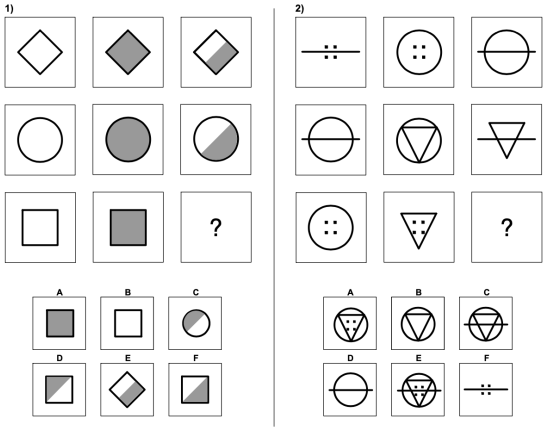
72

Cognitive Domains

Vocabulary

ZEALOT
BANAL
CATACOMB
GAOLED
FACADE
CELLIST
TOPIARY
QUADRUPED
ABSTEMIOUS
GAUCHE
AVER
DETENTE
IDYLL
RADIX
AEON
PUERPERAL
BEATIFY
LEVIATHAN

Reasoning



73

CLINICAL PSYCHOLOGY ASSESSMENT SUMMARY

Background

David suffered a head on collision with a car whilst cycling on his bicycle on 31/07/2015. He was admitted to the Royal Stoke University Hospital, where he received emergency neurosurgery. David was subsequently admitted to the Broadfield Ward on 25/09/2015 for a period of assessment and rehabilitation.

Reason for assessment

David was admitted to Broadfield as a polytrauma patient, having a severe traumatic head injury (extradural haematoma right side, focal contusions, base of skull fractures).

Considering the extent of his head injury, a request was made by the multidisciplinary team for a cognitive assessment to be carried out to assess David's cognitive functioning and neuropsychological status. Baseline screening measures of cognitive function indicated mild cognitive impairments.

With David's consent, the following assessments were carried out

- Neurocognitive testing:
  - Repeatable Battery for the Assessment of Neuropsychological Status (RBANS)
  - Test of Premorbid Functioning (ToPF)
  - Neurobehavioural Functioning Inventory (NFI)
  - Frontal Systems Behaviour Scale (FrSBSe)

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Presentation during assessment

During the assessment period David has displayed a high level of engagement with staff from all disciplines. David is able to understand and comply well with task instructions and has maintained a good level of effort during all the assessments.

David has shown a high degree of self-motivation and appears to enjoy challenges presented to him. During these he would verbalise sound problem-solving strategies used which reflected good deductive reasoning. This solution-focused and determined approach appears consistent with his premorbid personality. David is self-employed as an IT consultant and achieved a first in his degree in Engineering. David's partner, Ruth, confirmed that she has observed no noticeable personality changes in David. Both David and his partner reported that a relative area of difficulty for David is his short-term memory, which has been present since before the injury. This has been evident in David using a range of compensatory strategies to carry out domestic and work-related activities – examples include frequently depending on notes recorded on his phone and ipad to remember information.

Frontal Systems Behaviour Scale (FrsSBE)

Given the extent of David's injury to the frontal lobe of the brain, the Frontal Systems Behaviour Scale (FrsSBE) was administered to both himself and his partner. This assesses behavioural functioning specifically related to the frontal lobes and systems of the brain – namely, executive functioning, disinhibition and apathy.

75

Cognition

Initial screening of David's cognitive functioning, using the Montreal Cognitive Assessment (MoCA), revealed some mild cognitive impairment, particularly in relation to memory processing. A more comprehensive neuropsychological assessment was therefore carried out alongside an estimation of David's pre-morbid intellectual ability. This provides an estimated level of David's cognitive functioning prior to the injury, which enables comparisons to be made with his current performance.

Estimates of Pre-Morbid Intellectual Ability

An estimation of David's pre-morbid intellectual abilities was based on results gathered from the Test of Premorbid Functioning (ToPF) and also his educational history. The findings estimate that David previously functioned within the high average range of cognitive functioning.

Repeatable Battery for the Assessment of Neuropsychological Status (RBANS)

The RBANS measures cognitive functioning across the following domains:

- Immediate memory (measures initial encoding and learning of verbal information)
- Delayed memory (measures delayed recall and recognition for verbal and visual information, including retrieving information from long-term memory)
- Language (measures expressive language, fluent use of language)
- Attention (measures the ability to register simple auditory information, visual scanning and processing speed)
- Visuo-spatial/Constructional (measures ability to perceive and process visuo-spatial information)

76

David's scores for this assessment are recorded in Table 1 below:

Table 1
RBANS Assessment Results 2(/10/2015

	Immediate Memory	Delayed Memory	Language	Attention	Visuo-spatial
Index Score	76	64	80	94	112
Classification	Borderline	Extremely low	Borderline	Average	High average
Percentile rank	5	1	6	34	79

Scores on the RBANS fall within seven progressive categories as follows: *extremely low, borderline, low average, average, high average, superior,* and *very superior.*

The percentile rank refers to comparisons to a normative sample – that is how others of David's age, without a brain injury, would be expected to perform in these areas.

As depicted in the table above, David's scores for this assessment indicated that his current cognitive functioning for visuo-spatial/constructional and attention falls within *high average* and *average* ranges respectively. David scored within the *borderline* range for immediate memory and language whilst his scores for delayed memory fall with the *extremely low* range.

This suggests that David's brain injury has likely affected his cognitive functioning, particularly in relation to these latter three areas and especially for delayed memory. Whilst David has reported that he has always had a poor memory, even prior to the injury, the estimation of his premorbid functioning would predict higher levels than that which are recorded above - for both delayed and immediate memory as well as language and attention.

Further analysis of the subtests for the delayed memory domain suggest that David has a particular difficulty with long-term verbal memory encoding and retrieval – that is a high rate of forgetting information he has heard but a better chance of recalling information that is presented visually.

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In particular it was recommended that David do the following:

1. Use a diary/calendar to note down important dates and times (such as appointments).
2. Keep a diary or a small notepad with you (such as on your phone or ipad) when you are out to make a note of something important that needs to be recorded or remembered later.
3. Identify a single area in the house that is visible and easily accessible to keep relevant items together (such as your keys, ipad, etc).
4. Establish a structured routine for upcoming weeks which includes specific time for rest.
5. For each activity, do not get tempted to keep going until you become fatigued. Instead pace yourself with rest periods and slowly grade up your activity levels (functional, vocational and social).
6. Create a filing system for all correspondence, content of appointments, bills and letters. Use separate compartments or trays and label these clearly.
7. Simplify environmental demands when completing tasks and/or processing information; ensure that the environment is as uncluttered and distraction-free as possible.
8. When completing new, prolonged or more complex tasks:
   - 8.1. Allocate time beforehand to consider the task in detail, read carefully any relevant material and to plan a procedure.
   - 8.2. Try to structure the task and section it into manageable steps.
   - 8.3. Keep a record of these activities – write out clear goals you have set with each step clearly defined. Make a note next to each one regarding whether or not this was achieved, and record the time you were able spend on each activity before tiring.
   - 8.4. After completing the activity, take time to consider the successes of the process and, if relevant, how you might complete the task differently if repeated.

David's partner and others who support him are advised to support David with the above recommendations as well as the following:

9. When conversing with David, do not overload him with too much information at once; instead break information down into manageable chunks.
10. Be receptive to when David may not have remembered something and be prepared to repeat the information.
11. Encourage David to write down and record key points of the relevant information.
12. Give David multiple opportunities to learn new information; talk things through with him to ensure he understands the information and encourage him to repeat it back to support successful encoding.
13. Try to help David to recall information by providing some cues/prompts to see if he can then remember it. For example, give him some choices to choose from, provide David with the first letter of the word or provide a cue to the context of the answer. Also, try directing David to any of the

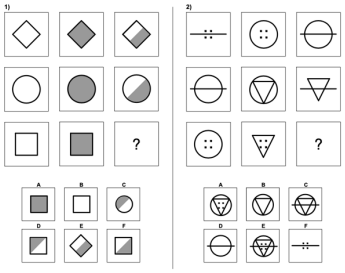
78

Cognitive domains

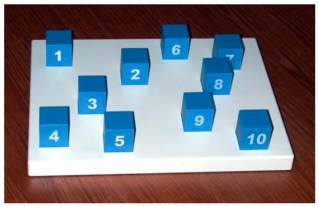
Vocabulary

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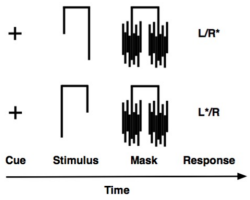
Reasoning



Memory



Processing Speed



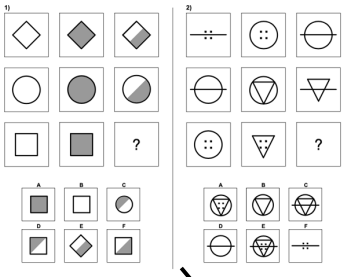
79

Cognitive domains

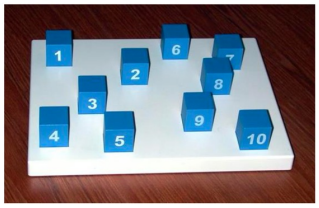
Vocabulary

ZEALOT
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FACADE
CELLIST
TOPIARY
QUADRUPED
ABSTEMIOUS
GAUCHE
AVER
DETENTE
IDYLL
RADIX
AEON
PUERPERAL
BEATIFY
LEVIATHAN

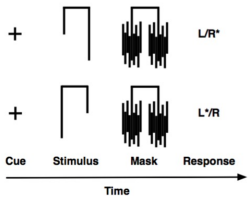
Reasoning

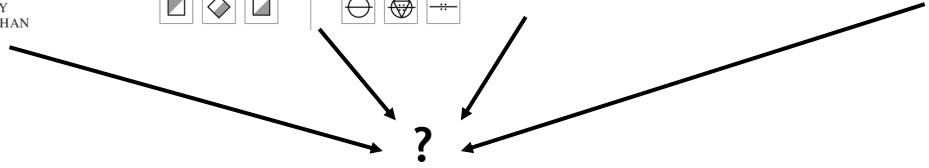


Memory



Processing Speed





80

g : general cognitive function
[much of our cognitive differences are general]



Ritchie et al. (2016) *Intelligence*, doi:10.1016/j.intell.2016.08.007

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